Tutorial-8

24/3/26

(Decision Tree)

Training Dataset :-

Email ID	suspicious words	money keywords	senders known	Has attachment	class
1	8	1	0	1	S
2	1	0	1	0	N
3	6	1	0	1	S
4	2	0	1	0	N
5	7	1	0	0	S
6	1	0	1	1	N
7	9	1	0	1	S
8	2	0	1	0	N
9	5	1	1	0	S
10	1	0	1	1	N

ID3 Algorithm (No Gini impurity)

↓

Entropy & Information Gain

1) which attribute should be selected as the root no
Justify your answer with calculation.

2) Build the complete Decision Tree using the ID3 alogorithm (use information Gain). Draw the tree structure with all nodes branches and leaf nodes clearly labeled.

3) what is the depth of your decision tree? How many leaf nodes does it have?

(IG) → Information Gain of each feature

IG → max → best feature

$$\boxed{\text{Goal :- Reduce Entropy} \quad \text{Increase Information Gain}}$$

$$\boxed{\text{Entropy} \rightarrow -\sum_{i=1}^{N} P_i * \log_2 (P_i)}$$

$$\log_2 P_i = \frac{\log_{10} P_i}{\log_{10} 2} = \frac{\ln P_i}{\ln 2}$$

IG = Entropy (parent)

$$\boxed{IG = -\sum_{k} \frac{N_k}{N} \text{Entropy}(k)}$$

1) Entropy (parent or y) =

$$P_{spam} = \frac{5}{10} = 0.5$$

$$P_{nspam} = \frac{5}{10} = 0.5$$

$$\text{Entropy} = -\left[0.5 * \log_2(0.5) + 0.5 * \log_2(0.5)\right]$$

$$= 1$$

⇒ Entropy (money Keywords)

1 → spam
→ non spam (x)

0 → non-spam
→ spam (x)

perfectly classified so Entropy = 0

$$IG(m.k) = 1 - 0 = 1$$

3) Entropy (sender known)

1 → spam (1)
→ non spam (5)

0 → spam (4)
→ non spam (0)

$$\text{Entropy} = -\left[\frac{1}{6}\log_2\left(\frac{1}{6}\right) + \frac{5}{6}\log_2\left(\frac{5}{6}\right) + 0 + 0 + \frac{4}{4}\log_2\left(\frac{4}{4}\right)\right]$$

for '1' →

for '0' →

$$= 0.650$$

$$\simeq 0.65$$

$$IG(S,K) = 1 - \left(\frac{6}{10} \times 0.65 \times \frac{4}{10} \times 0\right)$$

$$= 0.61$$

4) Entropy (Has Attachments) :-

0 — spam = 2, non-spam = 3 } Entropy > 0

1 — spam = 3, non-spam = 2 } Entropy > 0

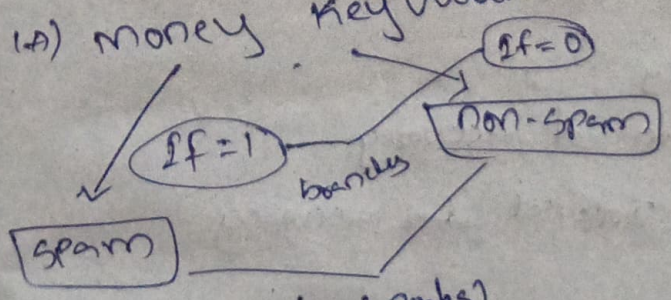
5) Entropy (suspicious words) :-

Entropy > 0 (Random value in dataset)

$\boxed{IG < 1}$

Root node

4) money Keywords :



(leaf nodes)

Depth = 1 (only one decision node is there)

Leaf nodes = 2